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Mid-Term Examination

Course: Artificial Intelligence (Machine Learning and Deep Learning
Time: 2 Hours 31-03-2026
Total Marks: 100

Name :- Ammar Course :- Artificial Intelligence

✅ Section A: Multiple Choice Questions (50 MCQs)

Highlight Yellow correct answer

◆ Data Types & Variables (Q1–Q10)

1. Which of the following is a mutable data type?
   a) tuple
   b) string
   c) list
   d) int
2. The data type of `10.5` is:
   a) int
   b) float
   c) complex
   d) str
3. Which function is used to check data type?
   a) type()
   b) check()
   c) typeof()
   d) datatype()
4. Variable names cannot start with:
   a) letter
   b) underscore
   c) number
   d) none

5. Which is correct variable assignment?
   a) x == 5
   b) x = 5
   c) 5 = x
   d) x := 5
6. Python is:
   a) statically typed
   b) dynamically typed
   c) strongly typed only
   d) none
7. Which data type stores key-value pairs?
   a) list
   b) tuple
   c) dictionary
   d) set
8. Output of `type(True)` is:
   a) int
   b) bool
   c) str
   d) float
9. Which is immutable?
   a) list
   b) dict
   c) string
   d) set
10. Which keyword defines a variable?
    a) var
    b) let
    c) none
    d) define

◆ Pandas & NumPy (Q11–Q20)

11. Library for numerical operations:
    a) pandas
    b) numpy
    c) matplotlib
    d) seaborn
12. Pandas main data structure:
    a) array
    b) dataframe
    c) matrix
    d) tensor

13. NumPy array is:
    a) heterogeneous
    b) homogeneous
    c) mixed
    d) none
14. Function to create array:
    a) np.create()
    b) np.array()
    c) np.make()
    d) np.build()
15. Pandas function to read CSV:
    a) read_file()
    b) read_csv()
    c) load_csv()
    d) import_csv()
16. Head function shows:
    a) last rows
    b) first rows
    c) all data
    d) summary
17. Shape returns:
    a) size
    b) rows & columns
    c) type
    d) index
18. NumPy stands for:
    a) Numerical Python
    b) New Python
    c) Numeric Processing
    d) none
19. Which is faster?
    a) list
    b) numpy array
    c) tuple
    d) dict
20. DataFrame is:
    a) 1D
    b) 2D
    c) 3D
    d) none

◆ Descriptive Statistics (Q21–Q30)

21. Mean is:
    a) average
    b) median
    c) mode
    d) range
22. Median is:
    a) middle value
    b) average
    c) sum
    d) none
23. Mode is:
    a) least frequent
    b) most frequent
    c) average
    d) none
24. Standard deviation measures:
    a) center
    b) spread
    c) count
    d) none
25. Variance is:
    a) sqrt of SD
    b) square of SD
    c) mean
    d) none
26. Pandas function for mean:
    a) avg()
    b) mean()
    c) median()
    d) sum()
27. Describe() gives:
    a) summary stats
    b) plot
    c) index
    d) none
28. Range =
    a) max - min
    b) mean
    c) variance
    d) none
29. Quartiles divide data into:
    a) 2
    b) 3
    c) 4
    d) 5

30. Skewness measures:
    a) symmetry
    b) center
    c) spread
    d) none

◆ Sampling & Population (Q31–Q35)

31. Population means:
    a) subset
    b) full dataset
    c) sample
    d) none
32. Sample is:
    a) whole data
    b) subset
    c) mean
    d) none
33. Random sampling ensures:
    a) bias
    b) fairness
    c) error
    d) none
34. Parameter relates to:
    a) sample
    b) population
    c) both
    d) none
35. Statistic relates to:
    a) population
    b) sample
    c) both
    d) none

◆ T-Test, Regression & Data Cleaning (Q36–Q50)

36. T-test compares:
    a) means
    b) variance
    c) sum
    d) none

37. p-value < 0.05 means:
    a) reject H0
    b) accept H0
    c) no change
    d) none
38. Null hypothesis means:
    a) difference exists
    b) no difference
    c) large difference
    d) none
39. Regression predicts:
    a) categories
    b) continuous values
    c) text
    d) none
40. Simple regression has:
    a) 1 predictor
    b) 2 predictors
    c) many
    d) none
41. Multiple regression has:
    a) one
    b) many predictors
    c) none
    d) fixed
42. Equation of regression:
    a) y = mx
    b) y = mx + c
    c) x+y
    d) none
43. R^2 measures:
    a) error
    b) model fit
    c) mean
    d) none
44. Residual =
    a) actual - predicted
    b) predicted - actual
    c) mean
    d) none
45. Linear regression assumes:
    a) non-linear
    b) linear relation
    c) random
    d) none

46. Data cleaning removes:
    a) noise
    b) data
    c) labels
    d) none
47. Missing values handled by:
    a) dropna()
    b) fillna()
    c) both
    d) none
48. Outliers affect:
    a) accuracy
    b) color
    c) size
    d) none
49. Which library for regression in Python?
    a) sklearn
    b) numpy
    c) pandas
    d) seaborn
50. Best fit line minimizes:
    a) error
    b) sum
    c) mean
    d) none

Section B: Short Questions (solve with Python)

(Each = 10 Marks)

◆ Q1: Data Types & Variables (Python Task)

Generate dataset and explain types:

```
import pandas as pd

data = {
    "Name": ["Ali", "Sara", "Ahmed"],
    "Age": [23, 25, 22],
    "Salary": [50000.5, 60000.75, 45000.25],
    "Employed": [True, True, False]
}
```

```
df = pd.DataFrame(data)

print(df.dtypes)
```

☞ Tasks:

- Identify data types
- Convert Age to float
- Add new variable Bonus

Answer :-

a) Identify data types :-

Code : -

print(df.dtypes)

Output :-

```
Name         object
Age           int64
Salary      float64
Employed       bool
```

b) Convert Age to float :-

Code :-

df["Age"] = df["Age"].astype(float)

print(df.dtypes)

Output :-

```
Name         object
Age         float64
Salary      float64
Employed       bool
```

c) **Add new variable Bonus :-**

Code :-

df["Bonus"] = df["Salary"]*1.5

print(df["Bonus"])

Output :-

```
0    75000.750
1    90001.125
2    67500.375
```

◆ Q2: T-Test (Hypothesis Testing)

```
import numpy as np
from scipy import stats

group1 = np.random.normal(50, 10, 30)
group2 = np.random.normal(55, 10, 30)

t_stat, p_val = stats.ttest_ind(group1, group2)

print("T-Statistic:", t_stat)
print("P-value:", p_val)
```

☞ Tasks:

- Define H0 and H1
- Interpret p-value
- Decision at 5% significance

Answer :-

a) **Define H0 and H1 :-**

- **H_0 (Null Hypothesis):** The statement that there is **no effect, no difference, or no relationship** in the population. It is what we assume to be true unless the data provides strong evidence against it.

  - Example: H_0: The average test score of students is 70.

- **H_1 (Alternative Hypothesis):** The statement that there **is an effect, a difference, or a relationship**. This is what we want to test or prove.

  - Example: H_1: The average test score of students is not 70.

b) **Interpret p-value :-**

```
print("P-value:", p_val)
P-value: 0.2259345166037645

As P-value is greater than 0.05 so means are non significantly different and
we accept H0
```

c) **Decesion at 5% significance :-**

If p-val > 0.05
- We accept H0
- Means are non significantly different

If p-val < 0.05
- We reject H0
- Means are significantly different

◆ Q3: Data Cleaning

```
import pandas as pd
import numpy as np

data = {
    "Age": [25, 30, None, 35, 40],
```

```
    "Salary": [50000, None, 60000, 65000, 70000],
    "City": ["Lahore", "Karachi", "Lahore", None, "Islamabad"]
}

df = pd.DataFrame(data)

print(df)
```

☞ Tasks:

- Handle missing values
- Remove duplicates
- Standardize text data
- Explain cleaning importance

Answer :-

a) Handle missing values :-

Code :-

df["Age"] = df["Age"].fillna(df["Age"].mean())

df["Salary"] = df["Salary"].fillna(df["Salary"].mean())

df["City"] = df["City"].fillna("Unknown")

Output :-

```
    Age   Salary       City
0  25.0  50000.0     Lahore
1  30.0  61250.0    Karachi
2  32.5  60000.0     Lahore
3  35.0  65000.0    Unknown
4  40.0  70000.0  Islamabad
```

b) Remove Duplicates :-

- **Code :-**

df = df.drop_duplicates()

- **Output :-**

```
    Age   Salary       City
0  25.0  50000.0     Lahore
1  30.0  61250.0    Karachi
2  32.5  60000.0     Lahore
3  35.0  65000.0    Unknown
4  40.0  70000.0  Islamabad
```

c) Standardize Text data :-

- **Code :-**

df["City"] = df["City"].str.lower()

df["City"] = df["City"].str.strip()

- **Output :-**

```
    Age   Salary       City
0  25.0  50000.0     lahore
1  30.0  61250.0    karachi
2  32.5  60000.0     lahore
3  35.0  65000.0    unknown
4  40.0  70000.0  islamabad
```

d) **Explain Cleaning Importance**

- **Improves Accuracy**
  a) Clean data ensures that analysis and results are correct and reliable.
  b) Errors, missing values, or duplicates can lead to wrong conclusions.
- **Enhances Consistency**
  a) Standardized and uniform data allows for meaningful comparisons and easier integration with other datasets.
- **Reduces Bias**
  a) Removing duplicates and correcting outliers prevents skewed or misleading results.
- **Saves Time and Resources**
  a) Clean data reduces the need for repeated checks and corrections during analysis or modeling.
- **Better Decision Making**
  a) Accurate and consistent data leads to informed and confident business or research decisions.
- **Improves Model Performance**

a) In machine learning, clean data helps models train better and make more accurate predictions.

◆ Q4: Regression Analysis

```
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression

X = np.array([1,2,3,4,5]).reshape(-1,1)
y = np.array([2,4,5,4,5])

model = LinearRegression()
model.fit(X, y)

print("Coefficient:", model.coef_)
print("Intercept:", model.intercept_)
```

☞ Tasks:

- Write regression equation
- Interpret coefficient
- Predict for X = 6
- Explain R^2 concept
- **Answer :-**

a) Write regression equation

- y=2.2+0.6·X

b) **Interpret Coefficient**

- • The coefficient **0.6** represents the **change in `y` for a one-unit increase in `x`**.
- • Interpretation: For every 1 unit increase in `x`, `y` increases by 0.6 on average.

c) Predict for X = 6

- Use the regression equation:
- y=2.2+0.6·6=2.2+3.6=5.8y = 2.2 + 0.6 \cdot 6 = 2.2 + 3.6 = 5.8y=2.2+0.6·6=2.2+3.6=5.8
- ✅ Predicted value: **y = 5.8**

d) **Explain R^2 Concept**

- R^2 (R-squared) measures **how well the regression model fits the data**.
- It is the proportion of the **variance in `y` explained by `x`**.
- **Value ranges from 0 to 1**:
  - 0 → model explains none of the variance
  - 1 → model explains all the variance
- Higher R^2 → better model fit.